FNP SALES ANALYSIS

Project Findings & Business Question Report

Objective
Explore sales performance, customer spending, product trends, occasion-based revenue, city-level orders, and delivery patterns using Excel.

Tools: Microsoft Excel · Power Query · Power Pivot · Data Model · PivotTables · Charts · Slicers · Timelines

Dataset: 100 customers · 1,000 orders · 70 products

Prepared from the project workbook and the supplied FNP business-question brief.

1. Executive Summary

This project analyses the supplied Ferns N Petals (FNP) sales dataset and presents the results through an interactive Excel dashboard. The workflow covered business-question framing, data inspection and preparation, data modelling, calculations, PivotTable analysis, visualisation, and interpretation.

The workbook reports total revenue of ■**35,20,984** from **1,000 orders**, an average delivery duration of **5.53 days**, and average revenue per order of approximately ■**3,520.98**. Monthly sales varied across the year, and the quantity-versus-delivery-duration correlation was approximately **0.003478**, indicating a negligible linear association.

2. Dataset and Analytical Process

The workbook contains three source tables: Customers (100 records), Orders (1,000 records), and Products (70 records). Power Query was used to inspect and prepare the data, including data-type adjustments, derived date/time and delivery-duration fields, and merging product price into the Orders data. Power Pivot relationships connected Customers and Products to Orders through their respective identifiers. Revenue was calculated from quantity multiplied by price. PivotTables then summarised the measures across time, product, category, city, and occasion.

3. Answers to the Ten Business Questions

1. Total Revenue

The dashboard KPI reports total revenue of ■**35,20,984** across **1,000 orders**. This provides the overall revenue baseline for the analysis.

2. Average Order and Delivery Time

The workbook reports an average delivery duration of **5.53 days**. This is the average difference between the recorded order and delivery dates. It can serve as a descriptive delivery-performance baseline; it does not by itself indicate whether delivery was on time against a promised service level.

3. Monthly Sales Performance (2023)

The monthly PivotTable shows substantial variation in revenue. The highest displayed months were **August: ■7,37,389** and **February: ■7,04,509**. March followed at ■5,11,823. The lowest displayed month was **January: ■95,468**. These results indicate uneven monthly sales in the workbook summary. Confirm the year/date filter in Excel before describing the values as a strictly 2023-only result.

4. Top Products by Revenue

The displayed top-five product summary is: **Magnam Set — ■1,21,905**; **Quia Gift — ■1,14,476**; **Dolores Gift — ■1,06,624**; **Harum Pack — ■1,01,556**; and **Deserunt Box — ■97,665**. Together, these five products contribute ■5,42,226 in the displayed summary.

5. Customer Spending Analysis

The KPI value is ■**3,520.984 per order**, calculated as total revenue (■35,20,984) divided by 1,000 orders. The correct metric name is **Average Order Value (AOV)**. It is not average spend per distinct customer unless the calculation uses the number of unique customers as its denominator.

6. Sales Performance by Top 5 Products

The PivotTable compares the revenue generated by the five leading products. Magnam Set is highest among the displayed five at ■1,21,905, followed by Quia Gift at ■1,14,476, Dolores Gift at ■1,06,624, Harum Pack at ■1,01,556, and Deserunt Box at ■97,665. This helps identify the strongest revenue contributors within the displayed top-five group.

7. Top 10 Cities by Number of Orders

The PivotTable identifies these ten cities and order counts: Imphal (29), Dhanbad (28), Kavali (27), Haridwar (24), Bidhannagar (21), Dibrugarh (21), Guntakal (20), North Dumdum (19), Bhatpara (18), and Bilaspur (18). These are the leading cities by order count in the displayed summary.

8. Order Quantity vs. Delivery Time

The calculated correlation coefficient is **0.003478** (approximately 0.0035). Correlation ranges from -1 to +1. A value this close to zero indicates virtually no linear association between order quantity and delivery duration in the analysed data. In this sample, higher quantities do not appear to be systematically associated with longer delivery durations. This does not prove there is no effect: correlation does not establish causation and does not rule out non-linear relationships or other influencing factors.

9. Revenue Comparison Between Occasions

The displayed occasion PivotTable reports: Anniversary — ■6,74,634; Raksha Bandhan — ■6,31,585; Holi — ■5,74,682; Birthday — ■4,08,194; Valentine's Day — ■3,31,930; and Diwali — ■3,13,783. Anniversary is the highest among the listed occasions. **Validation note:** these six values sum to ■29,34,808, which is ■5,86,176 below the overall revenue KPI. The displayed PivotTable may exclude records or categories, or have a filter applied. Reconcile its filters and included categories before treating it as a complete breakdown of total revenue.

10. Product Popularity by Occasion

The project used PivotTable and dashboard filtering to compare product popularity across occasions. The analysis indicated that the leading product changes with the occasion; therefore, no single product should be described as universally most popular for every occasion. Occasion-specific comparisons are more meaningful. The workbook summary available for this report does not expose a complete product-by-occasion numeric table, so no unsupported product names or figures are asserted here.

4. Additional Category Finding

The category PivotTable reports Colors at ■**10,05,645**, Soft Toys at ■**7,40,831**, Sweets at ■**7,33,842**, Cake at ■**3,29,862**, Plants at ■**2,12,281**, Mugs at ■**2,01,151**, and Raksha Bandhan at ■**2,97,372**. Note that the final label appears as a category in the workbook output; verify the source field/category mapping before using it as a product category in external reporting. The category totals displayed sum to the overall revenue KPI.

5. Dashboard and Interactivity

The dashboard includes KPI cards for total revenue, total orders, average delivery duration, and average order value. Charts cover occasion revenue, category revenue, top cities by order count, monthly revenue, top-five products by revenue, and revenue by order hour. Slicers and date timelines were connected to the relevant PivotTables; the project owner reports that the controls and chart updates were tested successfully.

6. Business Interpretation

The monthly results can help the business review periods of stronger and weaker revenue and investigate whether campaigns, occasion demand, or product availability contributed to the pattern. Product and category comparisons can inform merchandising discussions. City-level order concentration can guide regional review. Occasion-based product comparisons support more tailored assortments. The 5.53-day average delivery duration offers an operational baseline, while the near-zero quantity correlation suggests that order quantity alone is not a useful linear indicator of delivery duration in this dataset.

7. Recommendations

• Review the causes of the higher-revenue months and assess whether the pattern is repeatable before planning campaigns or inventory.
• Examine the leading products and categories for availability, margin, and promotion opportunities; revenue alone does not show profitability.
• Use city-level order counts to prioritise further regional analysis, while considering population, marketing exposure, and order value.
• Build occasion-specific product recommendations rather than assuming one product leads across all occasions.
• Investigate delivery duration alongside location, order date, product, and operational capacity; do not infer that quantity causes delivery delays from correlation alone.
• Reconcile the occasion PivotTable and verify the date scope of the monthly PivotTable before publishing those breakdowns as complete.

8. Limitations and Validation Notes

This is a descriptive analysis of the supplied workbook. The findings do not establish causality, explain why a pattern occurred, or predict future sales. The AOV metric is per order, not per unique customer. The displayed occasion totals do not reconcile to total revenue and require filter/category validation. The monthly chart's year scope should be checked against the assignment's 2023 requirement. The product-by-occasion numeric detail was not available in the workbook's extracted PivotTable summary, so the report records only the qualitative finding supplied during project discussion.

9. Conclusion

The project demonstrates a practical Excel-based analytics process, from translating a business brief into questions through data preparation, relational modelling, calculation, aggregation, dashboard development, and interpretation. The resulting workbook provides an interactive view of FNP sales and delivery patterns. The validation notes above should be resolved before treating every breakdown as a fully reconciled management report.

Source basis: supplied FNP business-question PDF, the uploaded Excel workbook (Project_1.xlsx), and project details confirmed in the discussion. Values are reproduced from the workbook's visible PivotTable/KPI outputs; caveats are included where those outputs do not reconcile or do not expose numeric detail.